scientific reports

OPEN

Influence of structural parameters on the workspace of delta parallel robots and path adaptability optimization for tea fresh leaf sorting applications

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In response to the requirements of "high speed, high precision, and low damage" in tea fresh leaf sorting using a Delta parallel robot, this study conducts workspace simulation based on the MATLAB platform. The objective is to determine the optimal reachable workspace under given structural parameter constraints, analyze the influence of structural parameters on the workspace, and optimize parameter combinations through orthogonal experiments to enhance the robot's adaptability to the sorting path (a gate-shaped trajectory). Firstly, the forward and inverse kinematic models of the Delta robot were established to derive the mapping relationship between the rotation angles of the active arms and the pose of the moving platform, thereby defining the mathematical representation of the workspace. Subsequently, key structural parameters—including the length of the active arms (L), the length of the passive arms (l), the circumradius of the base platform (R), and the circumradius of the moving platform (r)—were selected as influencing factors. An L9 orthogonal array was employed to design the experiments. The workspace volume for each parameter set was computed via MATLAB simulations, and the coverage of a gate-shaped trajectory (200 mm × 400 mm × 200 mm) was qualitatively assessed through visualization. Finally, the range analysis method was employed to determine the influence weight of each structural parameter on the workspace, leading to the identification of an optimal parameter set. The results demonstrated that the order of parameter influence was l > L > R > r. The optimized combination of the four parameters was obtained, which achieved complete coverage of the gate-shaped trajectory required for tea fresh leaf sorting. This study clarifies the quantitative relationship between structural parameters and the workspace, providing a scientific basis for the optimized design of Delta robots in tea leaf sorting applications. The proposed approach effectively addresses the long-standing issues of poor path adaptability and high damage rate of buds and leaves in conventional sorting robots.

Keywords Delta robot, Workspace, Kinematics, Orthogonal experiment, Parameter influence weight, Gate-shaped trajectory

The sorting of fresh tea leaves is a critical stage in the fine processing of tea, directly influencing the quality grade and added value of the final product. Traditional manual sorting methods suffer from high labor intensity, poor consistency, and low efficiency. In contrast, automated sorting technologies, with their advantages of high precision and efficiency, have increasingly become a research focus in the field of tea processing. Among these, Delta parallel robots, characterized by high stiffness, high speed, and high positioning accuracy, have demonstrated substantial potential in light industrial applications such as tea leaf sorting and packaging. However, existing studies have primarily focused on general material sorting tasks. Systematic analyses of the effects of Delta robot structural parameters on the workspace in the specific context of fresh tea leaf sorting remain limited, which constrains their practical application in precise design and task optimization.

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n the field of kinematic and workspace analysis of Delta robots, numerous studies have investigated their kinematic models and workspace characteristics. Hong et al.[1] proposed a workspace analysis method for Delta robots based on the concept of a zero platform, which employs geometric modeling and sectional visualization to simplify spatial computation under complex constraints, thereby providing a new perspective for subsequent parametric analyses. Cong et al.[2]integrated a vision system and implemented real-time sorting using a Delta robot based on the YOLOv1 algorithm. By applying a polynomial calibration method to optimize the coordinate mapping between the robot and the camera, they achieved a significant improvement in sorting accuracy. Guo et al.[3], focusing on tomato packaging scenarios, optimized the trajectory planning of Delta robots by adopting a 3-5-5-3 polynomial interpolation approach combined with an improved particle swarm optimization algorithm, effectively reducing rigidity-induced impacts and enhancing the stability of the sorting process. Zheng et al.[4] established a kinematic model of the Delta robot using a geometric approach and analyzed its workspace configuration through MATLAB simulations, providing a foundation for structural parameter optimization.

Although existing studies have provided important theoretical support for the application of Delta robots, most have focused on general material sorting tasks, with limited attention to the specific operational requirements of fresh tea leaf sorting. The sorting process of fresh tea leaves necessitates a gantry-type trajectory throughout the "grasp–lift–translate–place" operation. Unlike conventional material sorting, fresh tea leaf sorting requires the robot to operate within a relatively confined and complex workspace, where the configuration of structural parameters—such as the lengths of the upper and lower arms and the radii of the moving and fixed platforms—exerts a significant influence on the workspace geometry. Improper parameter configurations may lead to trajectory interference, vibration, or control instability, thereby compromising sorting accuracy and operational stability. Consequently, the optimization of Delta robot structural parameters for fresh tea leaf sorting—particularly the analysis of workspace–task path adaptability—has emerged as a critical issue requiring systematic investigation.

To address these challenges, this study investigates the kinematic analysis and workspace optimization of a Delta robot within the application context of fresh tea leaf sorting. First, a three-dimensional structural model of the Delta robot is established, and the position and orientation of the end-effector are derived through forward and inverse kinematic analysis. Considering the gantry-type trajectory requirements of the fresh tea leaf sorting process, the workspace of the Delta robot is subsequently determined. An orthogonal experimental design is then employed to systematically analyze the influence and relative significance of structural parameters on the workspace. Finally, based on the results of the above analyses, the optimal combination of structural parameters is obtained to enhance the adaptability of the Delta robot's workspace to the gantry trajectory of tea leaf sorting. This optimization effectively reduces rigidity-induced impacts and vibrations, ensuring smooth and stable operation during the sorting process.

Furthermore, to ensure that the robot meets the height requirements of the end-effector during the grasping and placing stages in the fresh tea leaf sorting process, the effects of structural parameters—including the lengths of the upper and lower arms as well as the radii of the moving and fixed platforms—on the horizontal workspace and width at a fixed height cross-section are further analyzed. Considering the spacing between the two conveyor belts and the corresponding operational range in the actual sorting environment, this study aims to achieve maximum task coverage by calculating and optimizing the robot's effective working width within the horizontal plane. Consequently, the optimal combination of structural parameters satisfying all physical and kinematic constraints is determined.

The main contributions of this study are as follows:

(1) A workspace optimization method for Delta robots specifically designed for fresh tea leaf sorting tasks is proposed, addressing the existing research gap in application-specific workspace optimization.
(2) Based on orthogonal experimental design and structural parameter sensitivity analysis, the influence of each parameter on the workspace is systematically evaluated, and corresponding optimization strategies are developed.
(3) Considering the practical task requirements, a structural optimization scheme based on maximum task coverage is proposed, providing a theoretical foundation for the design of Delta robots in automated tea leaf sorting.

The outcomes of this study offer significant theoretical support for the design and application of Delta robots in fresh tea leaf automated sorting, thereby promoting the advancement of tea processing toward greater intelligence and precision.

Kinematic and structural modeling of parallel robots

To permit unconstrained motion in three-dimensional space, a robot should possess at least six degrees of freedom (DOF). However, an increase in the number of degrees of freedom also results in greater computational and control complexity[5]. Therefore, it is essential to select a robot with the minimum necessary degrees of freedom while still fulfilling task requirements[6]. Typically, the degrees of freedom for parallel robots are calculated using the following formula:

$$F = d\,(n - g - 1) + \sum_{i=1}^{g} f_i \tag{1}$$

In automated production lines for fresh tea leaf sorting, an industrial robotic system with high-speed sorting capability is required to achieve the precise removal of coarse and aged leaf buds. In this study, a three-degree-of-freedom parallel robot based on a Delta mechanism is employed as the execution unit, with its kinematic characteristics illustrated in Fig. 1. The robot system consists of a fixed platform, a moving platform, and three symmetrically arranged active kinematic chains[7].

The static platform serves as the fixed reference plane and is rigidly connected to the system frame. It integrates three independent drive units, rotary-axis stepper motors, and a vision-based detection module. Each drive unit is composed of a 42-stepper motor and a gear transmission pair, which together control the coordinated motion of the three kinematic chains[8]. The spatial pose variation of the moving platform is achieved through the coupled kinematics of the three chains, while a pneumatically actuated suction-type end-effector mounted on the platform performs the target grasping tasks. An additional rotary degree of freedom is controlled by a stepper motor installed on the static platform, enabling precise adjustment of the end-effector's out-of-plane orientation[9].

As shown in Fig. 1, the Delta parallel robotic system adopts a three-degree-of-freedom spatial translational configuration. It is composed of a static platform, a moving platform, and three symmetrically distributed pairs of active and passive arms forming the kinematic chains. The static platform, serving as the primary load-bearing structure, is rigidly fixed and integrates three stepper motor drive units, which transmit output torque to actuate each kinematic chain.

Each kinematic chain transmits motion through two revolute joints: the first connects the motor output shaft to the active arm, while the second links the active arm to the passive arm. At the distal end of each passive arm, four spherical bearings are employed to form a spatial parallelogram mechanism, thereby establishing a closed-loop motion constraint. The moving platform, serving as the mounting base for the end-effector, has its pose governed by the coordinated control of the three kinematic chains, enabling decoupled translational motions along the X, Y, and Z axes within the Cartesian coordinate system.

Kinematic analysis and workspace determination

The kinematic analysis of the Delta robot can be classified into forward and inverse solutions. In the forward solution, the position of the end-effector's central point is determined when the rotation angles of the three driving joints are known[10], whereas in the inverse solution, the rotation angles of the three driving joints are calculated when the position of the end-effector's central point is given. A schematic diagram of the single-chain structure of the Delta robot is presented in Fig. 2:

Inverse kinematic analysis

To facilitate the formulation of the kinematic equations, one of the kinematic chains is selected for analysis, as illustrated in Fig. 2. The circumscribed circle radius of the equilateral triangle formed by the driving joints on the fixed platform is denoted as *R*, while that formed by the joints on the moving platform is denoted as *r* The length of the active link is represented by *L*, and the length of the passive link is represented by *l*. Accordingly, the position vector equation of the end-effector can be expressed as[11]:

$$\overrightarrow{OP} = \overrightarrow{OA_i} + \overrightarrow{AB_i} + \overrightarrow{BP_i} \tag{2}$$

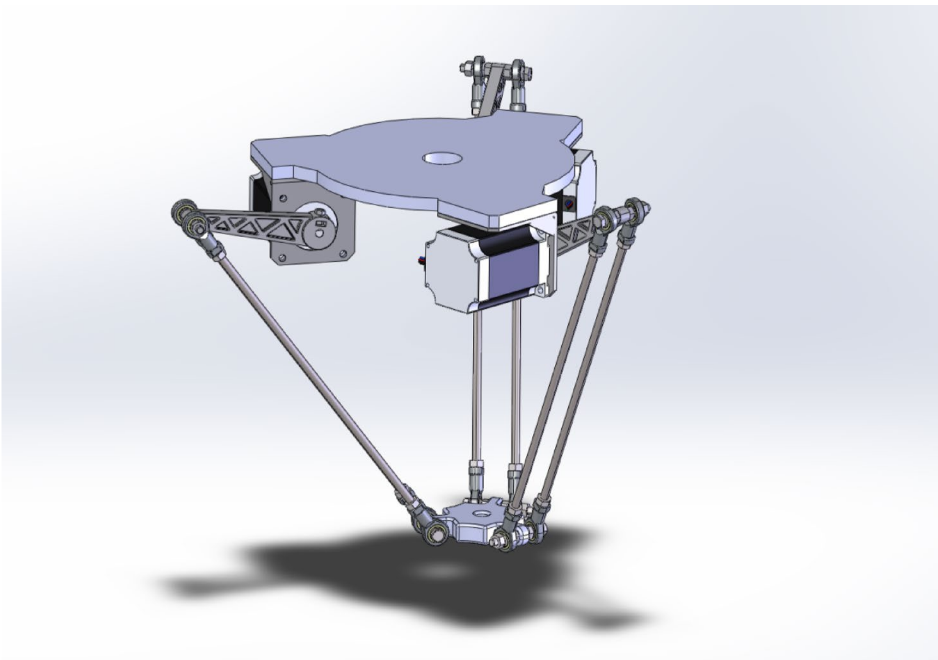

Fig. 1. Model of the parallel robot.

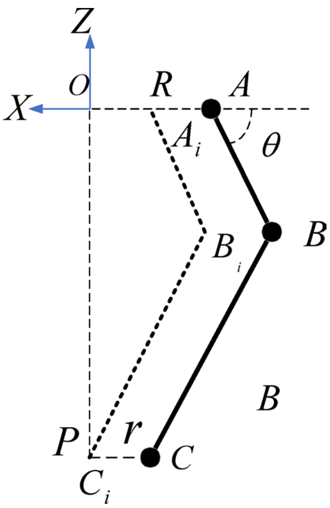

Fig. 2. Schematic representation of a single kinematic chain.

The inverse kinematic solution of the Delta parallel robot is relatively straightforward. The procedure is established on the geometric analysis of each kinematic chain with respect to its length and spatial configuration[11]. Based on these geometric relationships, the corresponding mathematical model is formulated, from which the numerical values of the joint angles are obtained[12]. The solution process is thus carried out through the analytical computation of the spatial geometric constraints imposed on each kinematic chain[13].

The points $B_i\ (i = 1, 2, 3)$are expressed in the$O - XYZ$coordinate frame as:

where $\alpha_i$denotes the azimuth angle between the vector $\overrightarrow{OA_i}$and the X-axis, defined by$\alpha_i = \frac{2\pi}{3}\,(i-1)\,(i = 1, 2, 3)$

$$\begin{bmatrix} x_i \\ y_i \\ z_i \end{bmatrix} = \begin{bmatrix} (R + L\cos\theta_i)\cos\alpha_i \\ (R + L\cos\theta i)\sin\alpha_i \\ -L\sin\theta_i \end{bmatrix} \tag{3}$$

The position of the point $C_i\ (i = 1, 2, 3)$ at the end of the moving arm in the$P - XYZ$ coordinate system is expressed as:

$$\begin{bmatrix} x_i \\ y_i \\ z_i \end{bmatrix} = \begin{bmatrix} r\cos\alpha_i \\ r\sin\alpha_i \\ 0 \end{bmatrix} \tag{4}$$

The center point P of the moving platform in the $O - XYZ$coordinate system is represented as $\begin{bmatrix} x & y & z \end{bmatrix}^T$ then the coordinates of $C_i\ (i = 1, 2, 3)$ in the $O - XYZ$ frame are given by:

$$\begin{bmatrix} x_i \\ y_i \\ z_i \end{bmatrix} = \begin{bmatrix} x + r\cos\alpha_i \\ y + r\sin\alpha_i \\ z \end{bmatrix} \tag{5}$$

Given that the fixed length of the active arm is denoted as $L = |\overrightarrow{A_iB_i}|$, and the fixed length of the moving arm is denoted as $l = |\overrightarrow{B_iC_i}|$,we can derive the following equation:

$$[(R - r + L\cos\theta_i)\cos\alpha_i - x]^2 + [(R - r + L\cos\theta_i)\sin\alpha_i - y]^2 + (L\sin\theta_i + z)^2 = l^2 \tag{6}$$

Using the trigonometric identities: $\sin\theta_i = \frac{2t}{1+t^2}$ and $\cos\theta_i = \frac{1-t^2}{1+t^2}$ the equation can be simplified as:

$$A_i t_i^2 + B_i t_i + C_i = 0 \tag{7}$$

$$\begin{cases} A_i = (R-r)^2 + x^2 + y^2 + z^2 + L^2 - l^2 + \\ \ 2\,(L - R + r)\,(x\cos\alpha_1 + y\sin\alpha_i) - 2L\,(R - r) \\ B_i = 4zL \\ C_i = (R-r)^2 + x^2 + y^2 + z^2 + L^2 - l^2 + \\ \ 2\,(r - L - R)\,(x\cos\alpha_i + y\sin\alpha_i) - 2L\,(R - r) \end{cases} \tag{8}$$

Where A_i、 B_i and C_i are known quantities, and the equation is a quadratic equation in t_i. Solving for t_i,we obtain:

$$t_i = \frac{-B_i \pm \sqrt{B_i^2 - 4A_iC_i}}{2A_i} \tag{9}$$

Therefore, when the coordinates of the center point of the moving platform in the $O - XYZ$reference frame are given, the three joint angles θ_i $(i = 1, 2, 3)$ can be obtained from the above Eq.

$$\theta_i = 2\arctan t_i \tag{10}$$

Forward kinematic analysis

For the Delta parallel robot mechanism, the forward kinematic analysis is carried out by defining the three primary driving joint angles. The goal is to determine the coordinates of the center point of the moving platform[13].

By substituting the values of $\alpha_i = \frac{2\pi}{3}(i-1)\,(i = 1, 2, 3)$ into Eq. (6), the system of equations can be obtained:

$$\begin{cases} \left[\frac{\sqrt{3}}{2}(R + L\sin\theta_1 - r) - x\right]^2 + \left[\frac{1}{2}(R + L\sin\theta_1 - r) - y\right]^2 + (L\sin\theta_1 + z)^2 = l^2 \\ \left[-\frac{\sqrt{3}}{2}(R + L\sin\theta_2 - r) - x\right]^2 + \left[\frac{1}{2}(R + L\sin\theta_2 - r) - y\right]^2 + (L\sin\theta_2 + z)^2 = l^2 \\ x^2 + [(R + L\sin\theta_3 - r) + y]^2 + (L\cos\theta_3 + z)^2 = l^2 \end{cases} \tag{11}$$

By simplifying Eq. (11), the following can be obtained:

$$\begin{cases} (A_1 - x)^2 + (B_1 - y)^2 + (C_1 + z)^2 = l^2 \\ (A_2 - x)^2 + (B_2 - y)^2 + (C_2 + z)^2 = l^2 \\ x^2 + (B_3 + y)^2 + (C_3 + z)^2 = l^2 \end{cases} \tag{12}$$

From the above equations, the following expressions can be derived:

$$A_1x + B_{13}y = C_{13}z + D_1 \tag{13}$$

$$A_2x + B_{23}y = C_{23}z + D_2 \tag{14}$$

$B_{13} = B_1 + B_3, B_{23} = B_2 + B_3, C_{13} = C_1 - C_3,$
$C_{23} = C_2 - C_3, D_1 = \frac{1}{2}(A_1^2 + B_1^2 + C_1^2 - B_3^2 - C_3^2), D_2 = \frac{1}{2}(A_2^2 + B_2^2 + C_2^2 - B_3^2 - C_3^2)$

From these, the expressions for *x* and *y* with respect to zzz are derived as follows:

$$\begin{cases} x = E_1z + F_1 \\ y = E_2z + F_2 \end{cases} \tag{15}$$

$E_1 = \frac{B_{13}C_{23} - B_{23}C_{13}}{A_2B_{13} - A_1B_{23}}, F_1 = \frac{B_{13}D_2 - B_{23}D_1}{A_2B_{13} - A_1B_{23}},$
$E_2 = \frac{A_2C_{13} - A_1C_{23}}{A_2B_{13} - A_1B_{23}}, F_2 = \frac{A_2D_1 - A_1D_2}{A_2B_{13} - A_1B_{23}}$

By substituting the above expressions into Eq. (12), a quadratic equation in *z* can be obtained:

$$az^2 + bz + c = 0 \tag{16}$$

$a = E_1^2 + E_2^2 + 1, b = 2E_1F_1 + 2E_2B_3 + 2E_2F_2 + 2C_3, c = F_1^2 + (B_3 + F_2)^2 + C_3^2 - l^2$

The expression for *z* can be derived as:

$$z = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \tag{17}$$

Upon analyzing the solutions, two sets of solutions are obtained. The first set corresponds to the lower surface of the moving platform (which satisfies the robot's constraints), while the second set corresponds to the upper surface of the moving platform (which does not satisfy the robot's constraints).

Therefore, the final expression for *z* is

$$z = \frac{-b - \sqrt{b^2 - 4ac}}{2a} \tag{18}$$

Workspace determination

The workspace of the Delta parallel robot is defined as the set of all possible positions that the end-effector can reach. It serves as a critical measure of the robot's performance and forms the foundation for robot path planning[14].

The factors influencing the robot's workspace include the dimensions of the mechanism, the limitations of the actuating joint angles, interference between the links, and the occurrence of singular configurations[15].

By solving the forward kinematics and comprehensively analyzing the robot's workspace, the relationship between the moving platform and the active arm angles can be established[16]. Therefore, by considering the factors influencing the robot's workspace and defining reasonable joint angle limits, the effective range of the moving platform's motion can be determined[17]. Taking into account the robot's structural parameters and operational conditions, the joint angles of the three active arms are constrained between − 15° and 90°, with the angle range divided into 100 increments. The robot's angular configuration is obtained through inverse kinematics, and the Delta robot's workspace is determined using MATLAB software. The computed workspace of the Delta parallel robot is illustrated in Fig. 3a–d, which represent the robot's three orthogonal views and isometric view, respectively[18].

Analysis of the operational workspace and gate-shaped path

Figure 4 illustrates the experimental Delta parallel robot prototype developed for the fresh tea leaf sorting task. The system is constructed using an aluminum alloy frame to ensure structural rigidity and lightweight characteristics. The upper and lower platforms are connected through three identical kinematic chains, each composed of an active upper arm driven by a servo motor and a passive parallelogram linkage ensuring the end-effector's translational motion. The end-effector is equipped with a pneumatic gripper designed to handle delicate tea leaves without mechanical damage. Two parallel conveyor belts are integrated beneath the workspace to simulate the dual-line sorting environment, providing an experimental basis for workspace analysis and trajectory planning under realistic operational conditions.

The workspace size of a Delta parallel robot is directly influenced by several structural parameters, including the rotational ranges of its active joints, the lengths of the passive arms, the dimensions of the moving platform, and the radius of the fixed base. By accurately visualizing the workspace, it becomes possible to optimize the robot's operational efficiency while avoiding mechanical interference. This geometric configuration is crucial for assessing the robot's practically attainable workspace. The structural model of the Delta parallel robot, shown in Fig. 5, illustrates its distinctive layout and mechanical elements, providing insight into its workspace characteristics.

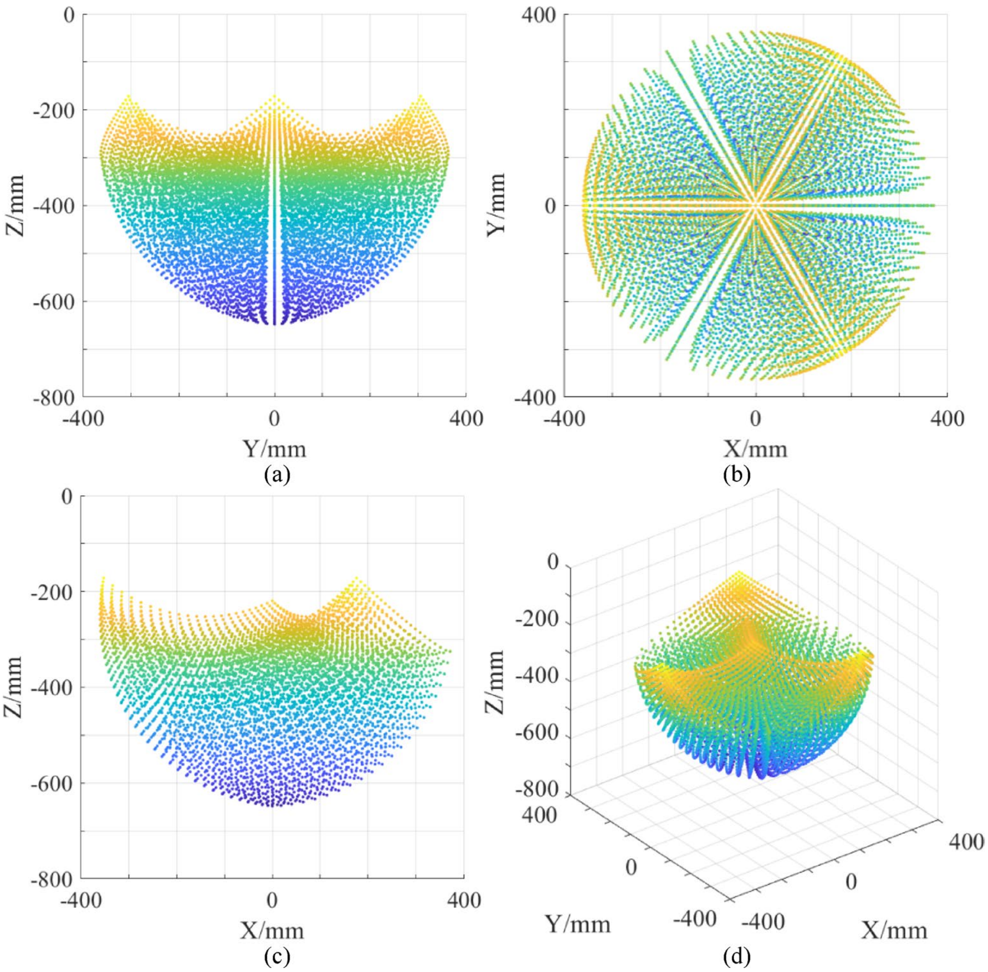

Fig. 3. Workspace of the delta parallel robot. (**a**) Front view, (**b**) Top view, (**c**) Side view, (**d**) Isometric view.

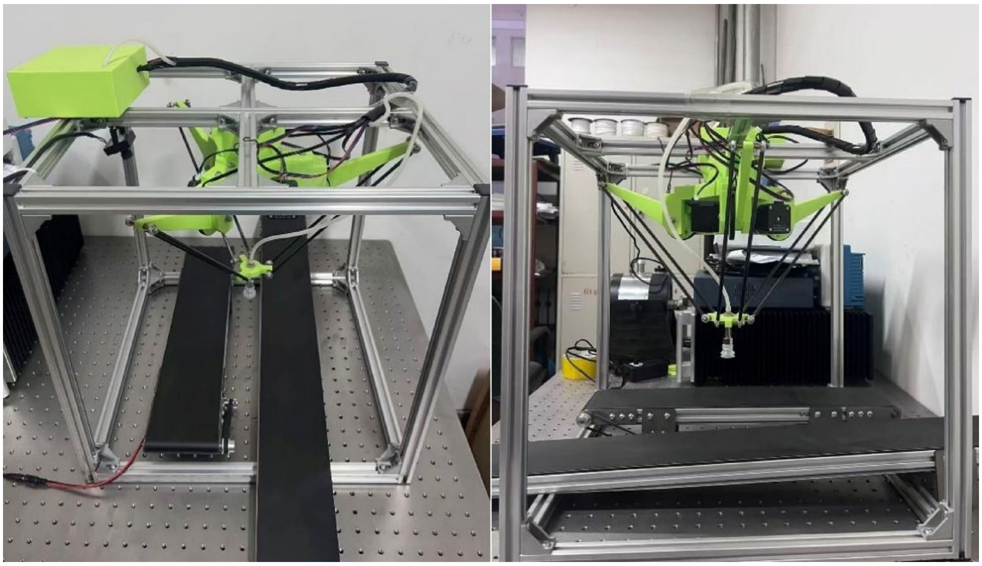

Fig. 4. Schematic diagram of the prototype structure.

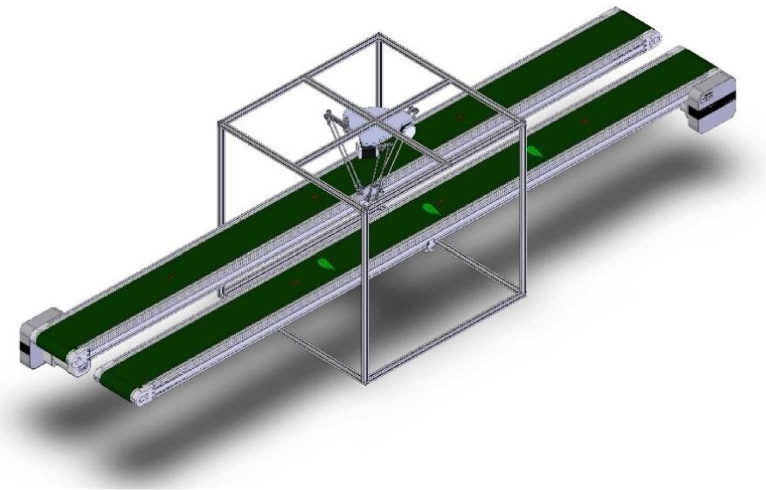

Fig. 5. Overall assembly diagram of the delta parallel robot.

As demonstrated in Fig. 6, the geometric extent and morphology of the workspace are determined by the robot's structural design and the kinematic constraints of its joints. This workspace is essential for the efficient execution of tasks, such as sorting coarse mature buds and leaves. To achieve high sorting accuracy and operational efficiency, the Delta parallel robot requires motion trajectory planning that aligns with the dimensional parameters of the production line equipment[14].

In the sorting application, the robot's motion is predominantly confined within a cylindrical workspace, as depicted in Fig. 7. The size and shape of this workspace are governed by the robot's structural configuration, ensuring that the motion stays within defined kinematic limits, thereby optimizing performance and minimizing potential interference.

Figure 7 illustrates the correlation between the robot's workspace and its actual operational trajectory, where the "gate-shaped trajectory path" defines the effective operational envelope during task execution. It is evident that the pick-and-place motions of the Delta parallel robot are predominantly confined within a relatively stable region of the workspace, bounded by the maximum reachable workspace and the realized motion trajectory. Consequently, when determining the optimal structural parameters, it is imperative to consider the spatial characteristics of the robot's workspace, so as to facilitate optimal trajectory planning and enhance the operational efficiency of leaf bud sorting in real-world applications.

By optimizing the workspace, the optimal structural parameters for tea leaf sorting can be identified based on the inherent geometric and kinematic constraints of the robot, including arm length, joint motion range, and platform dimensions. This optimization plays a crucial role in subsequent path planning and pick-and-

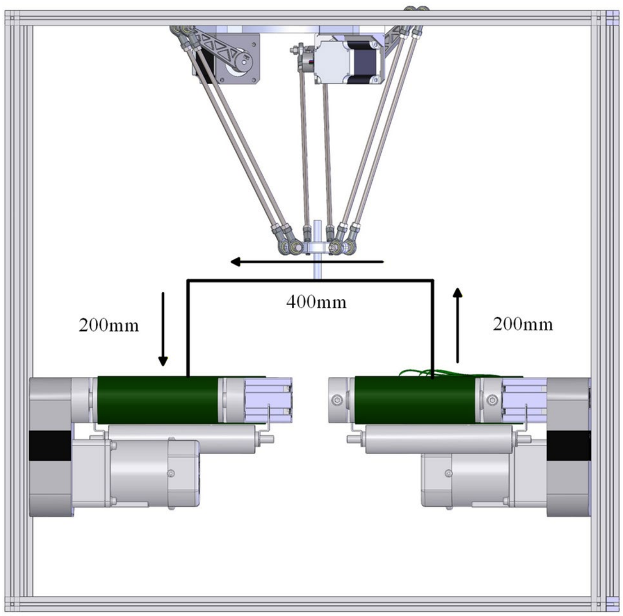

Fig. 6. Front view of the overall assembly of the delta parallel robot.

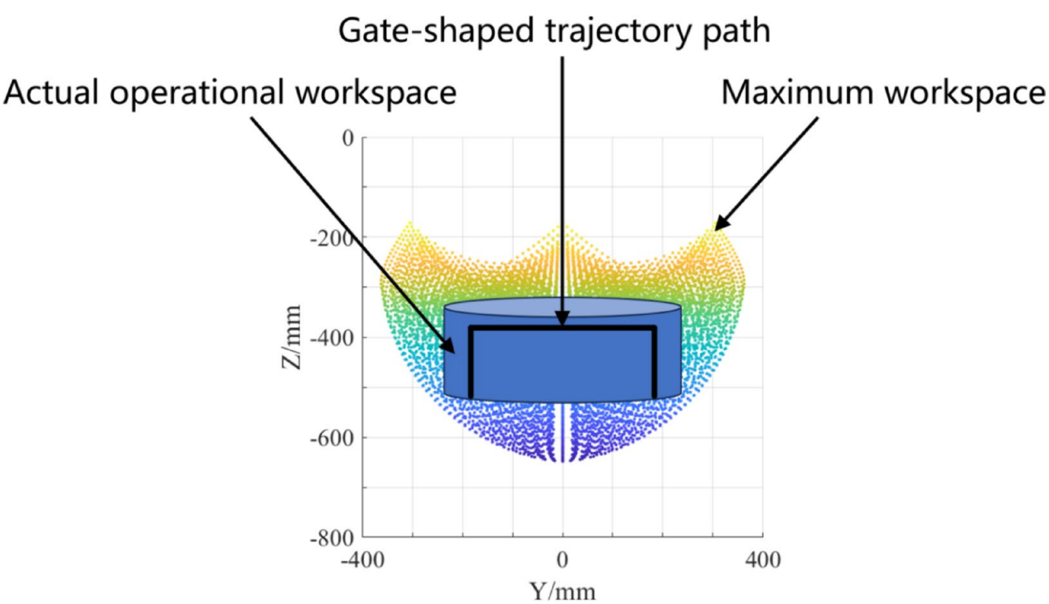

Fig. 7. Workspace and operational workspace of the delta parallel robot.

place trajectory refinement, thereby improving the robot's operational performance and efficiency in real-world sorting tasks.

Orthogonal experimental design for solving the optimal structural parameters of the delta parallel robot

The analysis of the Delta robot's workspace indicates that the primary factors influencing the workspace are the active arm length L, passive arm length l, the radius of the circumscribed circle of the static platform R, and the radius of the circumscribed circle of the moving platform r. In the field of industrial automation, particularly in the sorting of coarse and aged tea buds during tea processing, the Delta robot must perform trajectory planning based on the specific dimensions of the production line equipment. This characteristic dictates that the robot's sorting trajectory must be strictly confined within the boundaries of its workspace[19].

In the context of tea leaf sorting, the Delta robot is required to precisely identify and sort coarse, old leaf buds on the conveyor belt. During this process, the robot's motion trajectory must not only meet the sorting efficiency requirements but also adapt to the spatial layout of the tea production line. Therefore, under the given structural parameter constraints, determining the optimal workspace of the Delta robot holds significant engineering implications. This will provide crucial design parameters for subsequent sorting trajectory planning.

To validate the practical value of this theory, the present study will demonstrate, through a specific design case, how to determine the optimal workspace of the Delta robot within the constraints of structural parameters. The results of this research will be directly applicable to the design and optimization of automated tea leaf sorting production lines.

To verify the practical significance of this theory, the present study aims to illustrate, through a specific design case, the methodology for determining the optimal workspace of the Delta robot within the confines of structural parameter constraints. The findings of this study are expected to have direct applications in the design and optimization of automated tea leaf sorting production lines.

Design parameter requirements

This study employs an orthogonal experimental design to investigate the effects of four structural parameters—active arm length, passive arm length, dynamic platform radius, and static platform radius—on the workspace of the Delta robot. Under the constraints of the tea leaf sorting production line's spatial limitations and operational requirements, particularly concerning the efficient identification, precise localization, and rapid picking of coarse old leaf buds, the study quantitatively evaluates the comprehensive impact of key structural parameters on the robot's effective workspace. The optimal parameter combination is then identified based on these evaluations[15].

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Taking the workspace volume and the maximum cross-sectional area of the Delta robot as evaluation indices, four structural parameters (L, l, R, r) were selected as the principal factors. Drawing upon prior experience in coarse leaf and bud sorting, preliminary investigations, and considerations of manufacturing feasibility, three reasonable and distinguishable levels were assigned to each factor. Within the predefined ranges of structural parameters (Table 1), the active arm length (L) was constrained to 150–250 mm, the passive arm length (l) to 300–500 mm, the radius of the circumscribed circle of the moving platform (R) to 90–150 mm, and the radius of the circumscribed circle of the static platform (r) to 30–50 mm. Each factor was discretized into three equally spaced levels, yielding three experimental test points. The detailed factor-level configuration adopted in the orthogonal experimental design is presented in Table 1.

Analysis of the effects of structural parameters in the orthogonal experiment

In this study, the orthogonal experimental method was employed, with the workspace of the Delta robot taken as the evaluation criterion. Four key structural parameters, as listed in the table, were considered. An L9 orthogonal array was adopted to design the experimental scheme, resulting in a total of nine test groups. For each parameter combination, the workspace boundary was determined through forward kinematic analysis of the Delta robot, and the workspace volume was accurately calculated using the reachable point density method. The calculated workspace volumes for all experimental groups are summarized in Table 2. Range analysis and analysis of variance (ANOVA) were subsequently performed to evaluate the significance of the influence of each structural parameter on the workspace volume. On this basis, the contribution ratio and relative weight of each parameter were determined, and the optimal parameter combination was identified.

In the process of grasping and sorting coarse old leaves and buds, the Delta parallel robot primarily performs precise movements within its reachable workspace, from one point to another. This movement typically follows a gate-shaped trajectory, which consists of three phases: grasping, transporting, and placing. Throughout the entire operation, the end effector of the robot's moving platform is involved in the entire process, executing the grasping and placing actions. During the execution of the grasping and sorting tasks, the movement path of the Delta parallel robot must be confined within its three-dimensional reachable workspace. This path generally follows a gate-shaped trajectory, comprising three stages: grasping, transporting, and placing. In the complete workflow, the robot's end effector executes the grasping and placing actions, encompassing the entire process (Fig. 8).

Therefore, this study designs the specific structural parameters of the Delta robot to ensure that its workspace can accommodate the execution of the standard gate-shaped pick-and-place trajectory (200–400 mm–200 mm).

Kinematic structural parameters	L/mm	l/mm	R/mm	r/mm
1	150	300	90	30
2	200	400	120	40
3	250	500	150	50

Table 1. Structural parameters influencing the orthogonal experimental design.

Experiment group number	L/mm	l/mm	R/mm	r/mm	Workspace volume $/mm^3$
1	150	300	90	30	1.90×10^7
2	150	400	120	40	2.54×10^7
3	150	500	150	50	3.11×10^7
4	200	300	120	50	3.06×10^7
5	200	400	150	30	3.62×10^7
6	200	500	90	40	7.65×10^7
7	250	300	150	40	2.21×10^7
8	250	400	90	50	8.37×10^7
9	250	500	120	30	9.21×10^7

Table 2. Impact of orthogonal experimental parameters on the results.

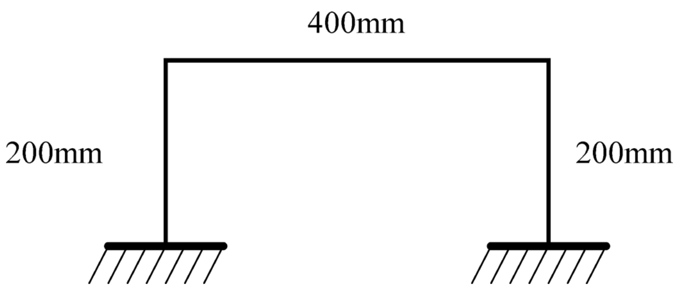

Fig. 8. End effector motion trajectory.

Based on the sorting requirements for fresh tea leaves, the experimental design specifies a grasping height that ascends from −500 mm to -300 mm. As shown in Fig. 8, the reasonable dimensions of the gate-shaped path suitable for experimental sorting are illustrated.[21]

To ensure that the Delta robot can efficiently perform the designated sorting and pick-and-place tasks, it is essential to rigorously verify that its workspace fully accommodates the standard gate-shaped trajectory. This necessitates precise structural parameter design to guarantee that the robot's workspace completely encompasses the predefined path, enabling the end-effector to execute grasping and placement operations within the specified height range. Therefore, this study optimizes key structural parameters of the Delta robot—such as the lengths of the upper and lower arms, and the radii of the moving and fixed platforms—to achieve an exact correspondence between the robot's workspace and the task trajectory.

To intuitively and accurately illustrate the morphological and dimensional variations of the Delta robot's workspace under different experimental conditions, a visualization cloud map approach, as shown in Fig. 9, was employed. This method enables a clear depiction of the geometric characteristics of the robot's operational region at a fixed height for each set of experimental parameters. The cloud map reveals the variations in the workspace corresponding to different structural configurations of the Delta robot, providing an intuitive understanding of the distribution and extent of its reachable area.

Furthermore, with the end-effector height strictly constrained during the pick-and-place operations, this study performs an in-depth analysis of how different structural parameters affect the robot's workspace, with particular attention to their influence on the horizontal workspace width at fixed height sections. By calculating the maximum attainable workspace width of the Delta robot under various combinations of structural parameters and sectional heights, the optimal structural configuration is identified. Building upon this, and taking into account the spacing and relative working ranges of the dual conveyor belts in the actual sorting process, an ideal parameter combination is further derived to achieve complete coverage of the task region and maximize operational effectiveness.

Based on the maximum horizontal cross-sectional analysis derived from the experimental results of Group 8, as shown in Fig. 10, it is evident that when the sectional height is$Z=-325mm$, the Delta robot's workspace achieves its most optimal horizontal reach, with an equivalent diameter of approximately 716.61mm and a maximum cross-sectional area of 403,321.44mm^2. This section not only represents the end-effector's optimal horizontal operational capability under vertical constraints but also reflects the overall impact of structural parameter configurations on the robot's accessible workspace. By examining the sectional shape variations across different structural parameter combinations, it becomes apparent that the linkage length, base radius, and moving platform radius play a significant role in shaping the upper region of the workspace. In particular, an increase in the base radius significantly expands the upper cross-sectional diameter, while the moving platform radius, while maintaining motion accessibility, limits the available workspace width.

Additionally, a comparative analysis between the spatial configuration at the maximum cross-section and the dual conveyor belt arrangement required for the sorting task demonstrates that by constraining the operational space to the vicinity of this section, it is possible to achieve full coverage of the operational area and

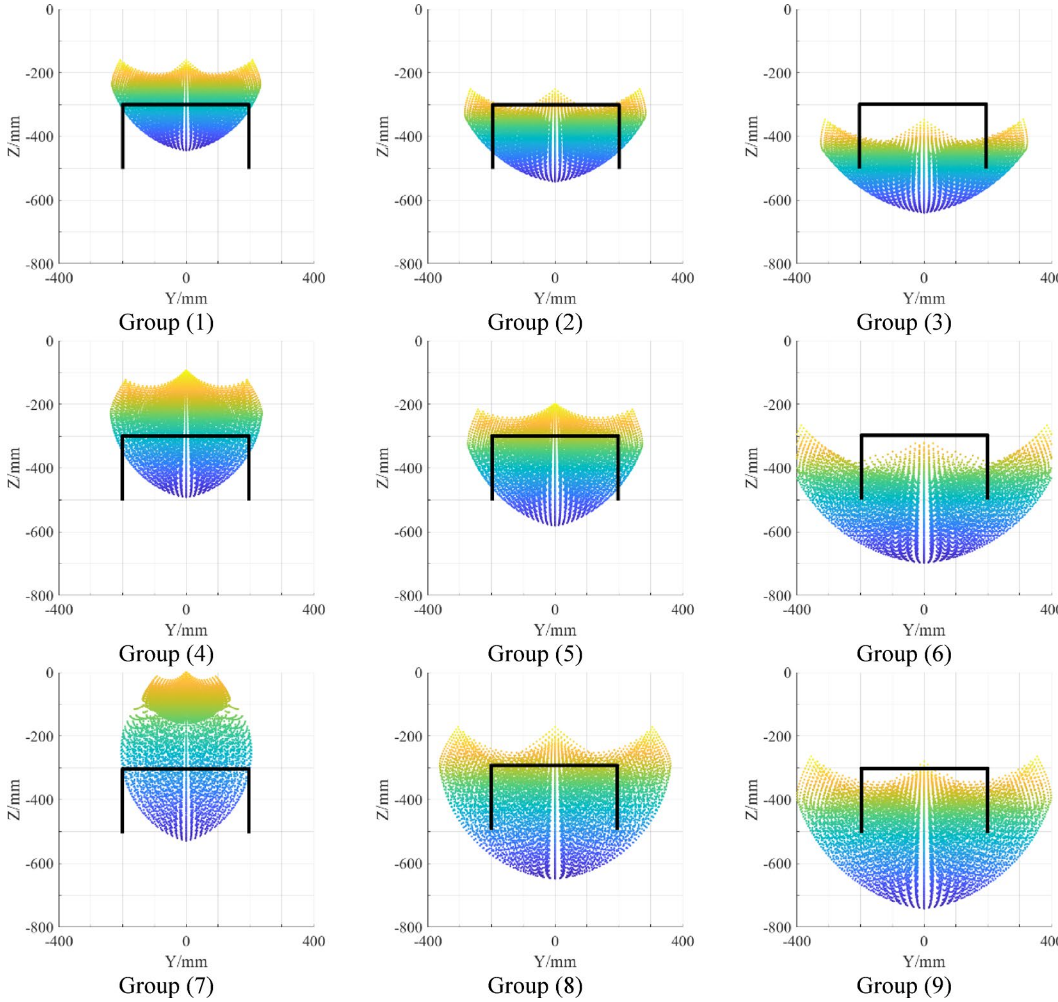

Fig. 9. Cloud map of workspace results for each experimental group.

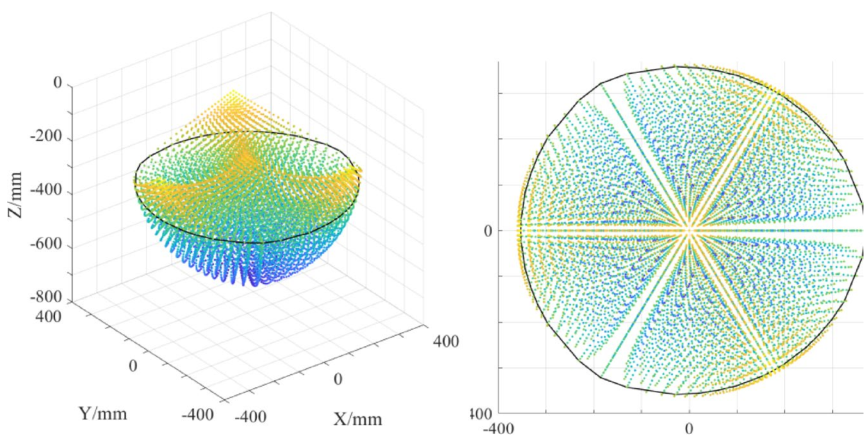

Fig. 10. Location of the maximum cross-sectional area.

optimal task reach while ensuring precise control of the end-effector height. Thus, under the constraint of a fixed end-effector height, this study integrates the cross-sectional maximization criterion with the task coverage optimization criterion to determine the design parameter range that ensures both maximal task coverage and optimal structural performance.

Ultimately, this study identifies the maximum achievable task coverage of the Delta robot while satisfying all physical and kinematic constraints. In addition, the maximum horizontal cross-sectional workspace areas under different structural parameter configurations are quantified. These results offer valuable insights and serve as a significant reference for the structural optimization and practical deployment of Delta robots in industrial sorting applications.

The workspace volume distribution for the nine experimental groups ranges from 1.90×10^7 mm³ to 9.21×10^7 mm³, with a range of 7.31×10^7 mm³. This indicates a systematic influence of the geometric parameter combinations on the workspace volume.

The results indicate that the length of the passive arm has the most significant impact on the workspace volume (range: 4.27×10^7 mm³). As the passive arm length increases from 300 mm to 500 mm, the average volume increases from 2.39×10^7 mm³ to 6.66×10^7 mm³, due to the extended reachable range of the moving platform with the longer passive arm. The length of the active arm has the second most significant effect (range: 4.07×10^7 mm³). When the active arm length is 250 mm, the average volume (6.59×10^7 mm³) increases 1.62 times compared to when it is 150 mm (2.52×10^7 mm³), suggesting that the increase in active arm length directly enhances the motion redundancy of the moving platform. The effect of the base radius (R) is moderate (range: 2.99×10^7 mm³). When the base radius is 90 mm, the average volume (5.97×10^7 mm³) is significantly greater than when it is 150 mm (2.98×10^7 mm³), as the increase in the base radius leads to greater interference between the active arm and the base, restricting the workspace expansion. The effect of the moving platform radius is minimal (range: 0.78×10^7 mm³). When the moving platform radius is 30 mm, the average volume (4.91×10^7 mm³) increases by 18.9% compared to when the radius is 40 mm (4.13×10^7 mm³). A smaller moving platform improves spatial utilization by reducing the constraints on the passive arm.

In the experiment, the gate-shaped trajectory is defined as a fixed rectangular boundary within the Y-Z plane (the black box in Fig. 6), with the requirement that the workspace fully covers this area to meet the motion constraints for precise pick-and-place operations. However, only the eighth group (with parameters: active arm length = 250 mm, passive arm length = 400 mm, base radius = 90 mm, and moving platform radius = 50 mm) effectively covers the gate-shaped trajectory region. As shown in Fig. 6, the workspace cloud map for the eighth group shows coverage, while the point clouds within the rectangular boundaries for the other groups (Groups 1, 2, 3, 4, 5, 6, 7, and 9) are sparse or missing, failing to meet the motion requirements of the gate-shaped trajectory.

An analysis of the eighth group's parameter combination reveals that the smaller base radius (90 mm) reduces interference between the active arm and the base, leaving sufficient space for the active arm to swing within the gate-shaped trajectory region. The moderate passive arm length (400 mm) strikes a balance between the travel distance and motion accuracy of the moving platform, preventing end effector vibrations caused by an excessively long passive arm. Although the larger moving platform radius (50 mm) slightly restricts the overall workspace volume (8.37×10^7 mm³), its compatibility with the 250 mm active arm ensures that the gate-shaped trajectory region (fixed Y range and Z height) remains within the effective workspace.

In contrast, the ninth group (with parameters: active arm length = 250 mm, passive arm length = 500 mm, base radius = 120 mm, and moving platform radius = 30 mm) has the largest workspace volume (9.21×10^7 mm³), but the larger base radius (120 mm) limits the movement range of the active arm, preventing coverage of the gate-shaped trajectory region. Additionally, the first group (with parameters: active arm length = 150 mm, passive arm length = 300 mm, base radius = 90 mm, and moving platform radius = 30 mm) has the smallest workspace volume and fails to cover the gate-shaped trajectory due to the small lengths of both the active and passive arms.

Conclusion

This study addresses the "high-speed, high-precision, and low-damage" requirements of the Delta parallel robot in the sorting operation of fresh tea leaves. A systematic investigation of kinematic analysis, workspace determination, and structural parameter optimization was conducted to reveal the influence of structural parameters on the workspace, providing a theoretical foundation for the design and application of automated sorting robots for fresh tea leaves. The forward and inverse kinematic models of the Delta robot were established, and through MATLAB simulation calculations, the three-view representation of the robot's workspace was obtained, visually demonstrating its spatial configuration. The influence weights of the Delta robot's structural parameters on the workspace are ranked as follows: passive arm length l> active arm length L> base radius R> moving platform radius r.

During the optimization design process, the advantages and disadvantages of each parameter were balanced according to the application scenario. In consideration of the "gate-shaped trajectory" (grasping-translation-placing) requirement for fresh tea leaf sorting, the optimal parameter combination that satisfies the trajectory adaptability was selected through workspace visualization evaluation. The selected parameters, such as those in Group 8 (active arm length = 250 mm, passive arm length = 400 mm, base radius = 90 mm, moving platform radius = 50 mm), achieved the optimal balance between workspace and motion performance.

Data availability

All data generated or analysed during this study are included in this published article.

Received: 27 September 2025; Accepted: 9 January 2026
Published online: 29 January 2026

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Acknowledgements
The authors appreciate the reviewers' valuable comments and suggestions. We would like to thank Editage (www.editage.cn) for the English language editing.

Author contributions
Chuanming Ren designed the innovative aspects and experimental process, independently performed all experiments, conducted data analysis, and drafted the manuscript. Wenguang Zheng supervised the project and revised the manuscript. Rongyang Wang collected experimental materials and assisted in manuscript revision. All authors approved the final manuscript.

Funding
This work was supported by the Huzhou Public Welfare Application Research Project (2024GZ26).

Declarations

Competing interests
The authors declare no competing interests.

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